WIRELESS SENSOR NETWORKS LABORATORY

Interactive Human-in-the-Loop Sorting System over a Self-Healing Wireless Sensor Network

7 nRF52840 nodes · Contiki-NG · NullNet + CSMA/CA · Bellman-Ford multi-hop routing

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Project Objectives

Bridging embedded sensing, operator judgment, and self-healing multi-hop networking

Resource Constraints

Achieving reliable multi-hop routing and telemetro on memory-constrained nRF52840 nodes.

The Interface Gap

Reconciling high-speed network packet routing with slow, asynchronous human decision-making

Physical Volatility

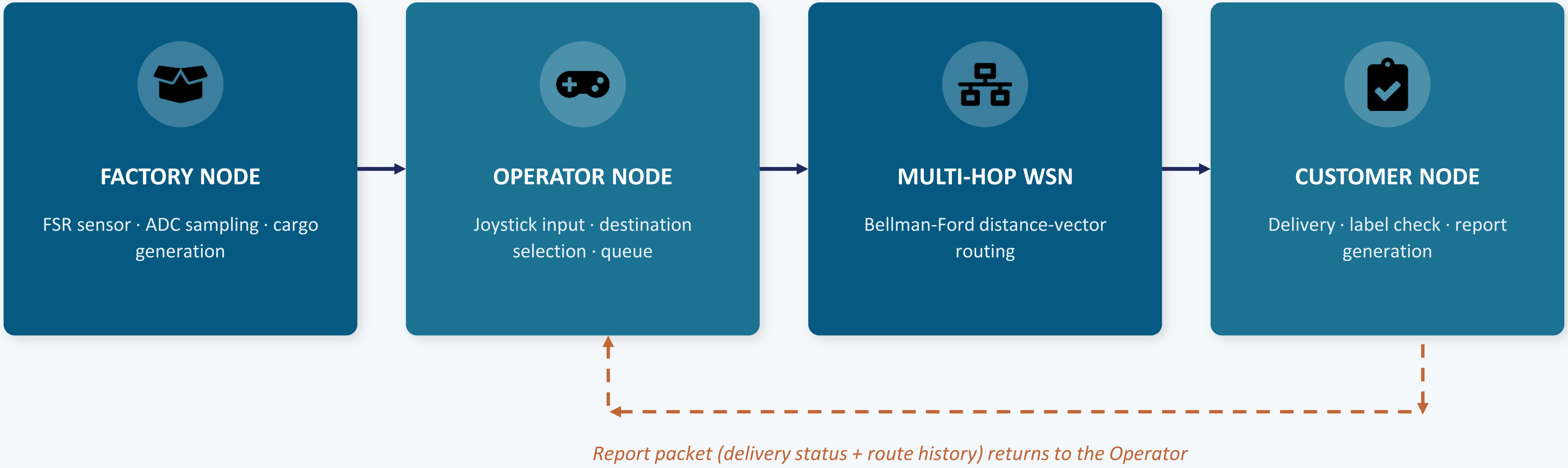
Maintaining an unbroken industrial sorting pipeline despite node failures and dynamic topology changes.

Self-Healing Delivery

Dynamic proxy routing keeps cargo flowing even when nodes fail.

System Architecture

Cargo flows from physical sensing through the network to the customer, with reports bouncing back



Stack: NullNet over IEEE 802.15.4 CSMA/CA, Channel 16 · Reduced TX power forces genuine multi-hop paths

Seven Nodes, Four Roles

Every node runs Contiki-NG; every node also participates in routing as a potential relay

Node 7

Factory

Polls FSR via ADC; generates cargo packet with true label on threshold + Button1 press.

Node 1

Operator

Buffers cargo (20-slot queue); Maps joystick X/Y to destination; Acts as the network bridge; Monitors delivery reports.

Nodes 2–5

Customers

Default delivery destinations; verify selected label vs. ground truth; generate reports.

Node 6

Dedicated Relay

Always available as intermediate hop; can also serve as a proxy customer.

Any node — including the Factory and Customers — can relay a packet destined for another device, enabling maximum physical flexibility.

Human-in-the-Loop Workflow

From physical force reading to verified delivery, in one continuous cycle

1

Cargo Generation

FSR threshold + Button1 press creates a cargo packet with a true label (A–D).

2

Operator Interaction

20-slot circular buffer queues cargo; joystick X/Y select the destination label.

3

Delivery Feedback

Customer node checks selected vs. true label; report bounces back via routing table.

4

Resolve or Re-sort

Correct → logged resolved.
Incorrect → re-queued at the back for a re-attempt.
Didn’t reach the node → Dropped.

Timeout while awaiting a report → cargo is dropped and logged as a failure.

Fault Tolerance & Self-Healing

Two mechanisms keep cargo moving even as the physical network changes

Failure Detection

- Beacons expose per-destination liveness; each node timestamps the latest beacon received.
- A watchdog checks these timestamps continuously.
- No update within a 10-second window → node marked offline.
- Routing cost to that node is set to infinity, so packets stop being forwarded through it.

Dynamic Proxy Routing

- Operator continuously monitors reachability of primary customers (Nodes 2–5).
- If one becomes unreachable, the routing table is checked for an available backup.
- Node 6 (relay) or even Node 7 (Factory) can step in as a proxy customer.
- Joystick input remaps to the proxy automatically — no manual reconfiguration or reset.

Once the original node recovers, its primary role is automatically restored.

Custom Packet Structure over NullNet

net_packet_t header (type, src_id) with a payload union: Beacon, Cargo, or Report

net_packet_t

type · src_id · payload

Beacon Payload

- Costs to known nodes

Cargo Payload

- Packet ID
- Destination ID
- Force value
- True label
- Selected label
- Route history
- Hop count

Report Payload

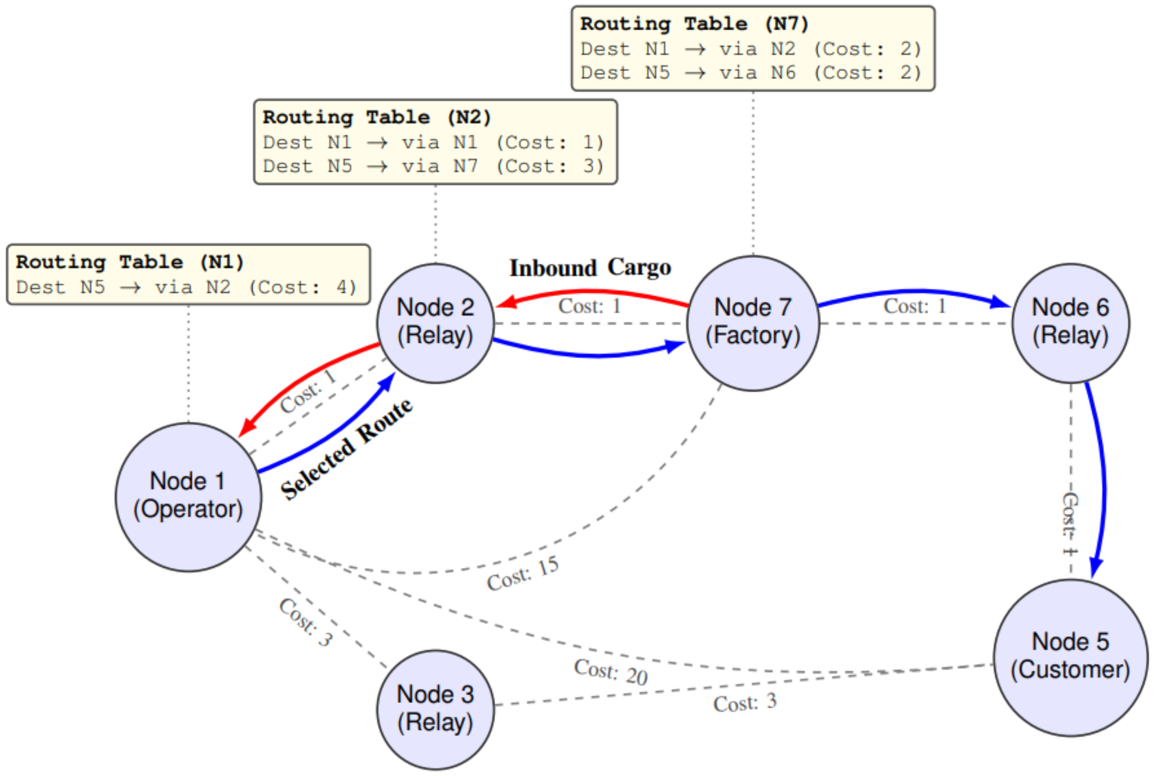
- Cargo ID
- Action
- Match success
- Route history
- Hop count

Multi-Hop Routing: Distributed Bellman-Ford

How it works

- Beacons broadcast every 3 seconds advertise routing costs to neighbors.
- RSSI on each received beacon is converted into strict penalty tiers (Cost:1, 4, 15, 50).
- Each node picks its next hop independently from its local table — no central coordination.
- Reduced TX power forces genuine multi-hop paths in the lab space.

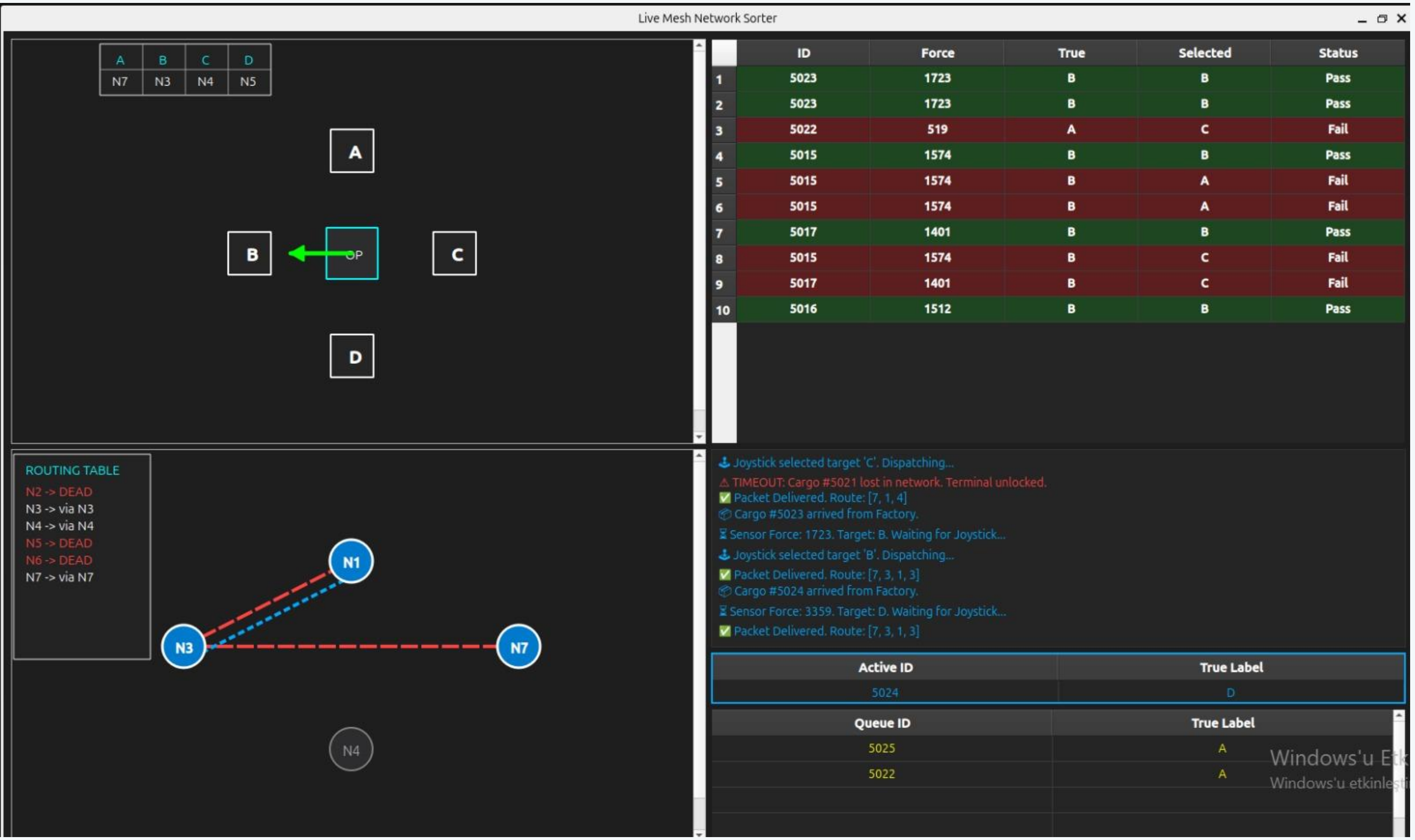
Example: Optimal Route via an Alternate Relay



Highlighted path shows a Bellman-Ford route using the Factory node as an intermediate relay.

Real-Time PyQt5 Dashboard

Serial link to the Operator node feeds a live visualization of network and delivery state



Status HUD

Maps customer labels A–D to physical nodes; updates live when proxy routing engages.

2D Topology View

Renders Bellman-Ford state and traces active cargo paths in real time.

Fault Color-Coding

Offline or dead nodes are flagged instantly for diagnosis.

Interactive Tables

Show the live circular buffer queue and historical sorting performance.

System Demonstration

Validated end-to-end on a physical network of 7 nRF52840 wireless sensor nodes

Workflow Verified

- Force sensor acquisition and cargo generation
- Packet buffering and joystick-based destination selection
- Forced multi-hop packet forwarding (lowered TX power)
- Accurate delivery feedback and reporting

Fault-Tolerance Validated

Node failures were intentionally introduced during the demo.

- Unreachable nodes were detected within the 10-second timeout.
- Dynamic proxy routing immediately reassigned unavailable customers.
- Cargo delivery continued without manual reconfiguration or software resets.

Conclusion

We designed and implemented a human-in-the-loop cargo sorting system that integrates embedded sensing, operator interaction, and distributed multi-hop communication on 7 nRF52840 nodes running Contiki-NG.

Custom packet structures

Beacon, Cargo, and Report payloads over NullNet

Distributed distance-vector routing

Bellman-Ford with RSSI-based link cost

Circular packet buffering

20-slot operator queue with re-sorting on failure

Dynamic fault recovery

Proxy routing keeps delivery alive despite node loss

Result: reliable, adaptive cargo delivery — with a human squarely in the loop.